Chapter 36

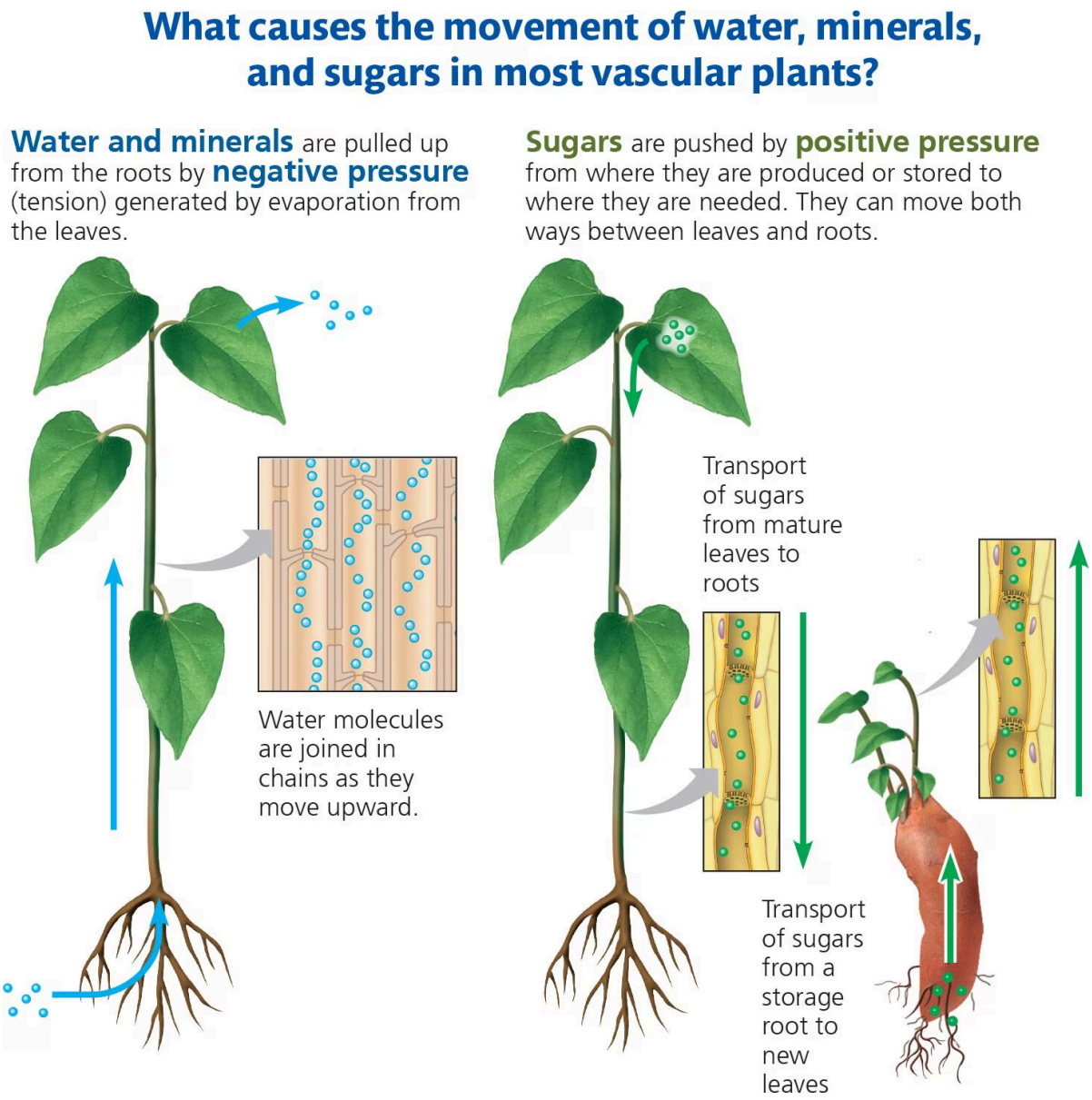
Campbell Biology, AP Edition, 12e

Chapter 36

Resource Acquisition and Transport in Vascular Plants

Figure 36.1

This English ivy has covered every square centimeter of a wall with foliage, an effective way to acquire light energy for photosynthesis. The sugars produced by the leaves and the water and minerals absorbed by the roots have to be transported throughout the plant.



Water and minerals are pulled up from the roots by negative pressure (tension) generated by evaporation from the leaves. Water molecules are joined in chains as they move upward. This is depicted by blue arrows on a plant moving upwards from the roots, through the stem, and to the leaves. An enlarged portion of the stem shows water molecules as droplets, moving in chains through the xylem.

Sugars are pushed by positive pressure from where they are produced or stored to where they are needed. They can move both ways between leaves and roots. In two examples, we see the transport of sugars from mature leaves to roots and the transport of sugars from storage roots to new leaves. This is depicted by green arrows going downwards from a leaf to the rest of the plant and upwards from storage roots to growing parts of the shoot, such as new leaves.

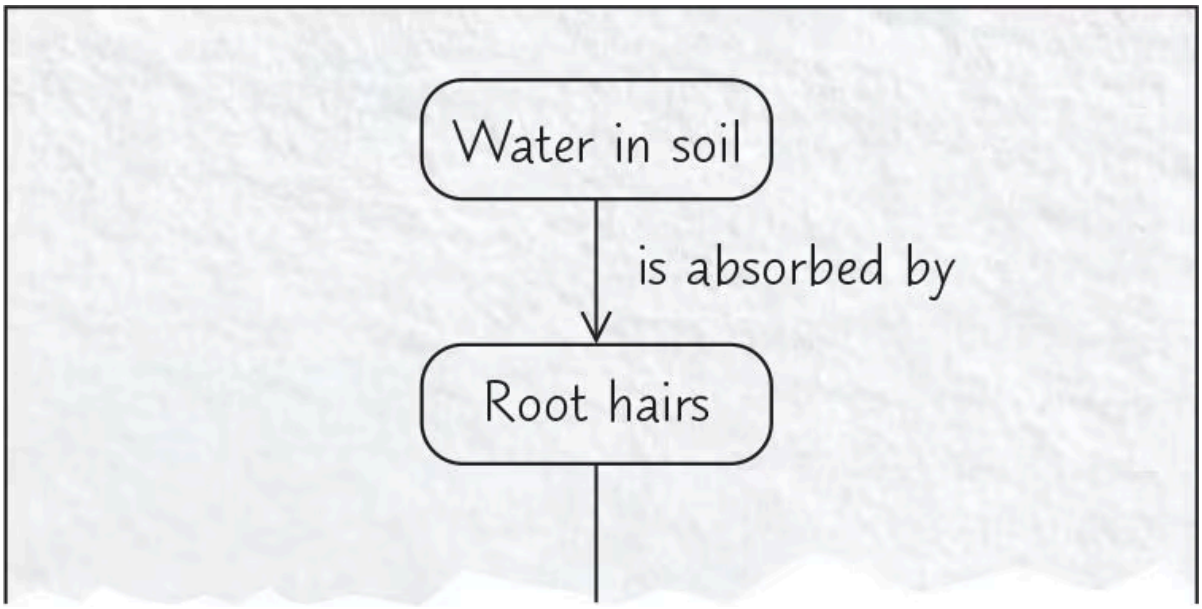
Key Concepts

- **36.1Adaptations for acquiring resources were key steps in the evolution of vascular plants****
- **36.2Different mechanisms transport substances over short or long distances****
- **36.3Transpiration drives the transport of water and minerals from roots to shoots via the xylem****
- **36.4The rate of transpiration is regulated by stomata****
- **36.5Sugars are transported from sources to sinks via the phloem****
- **36.6The symplast is highly dynamic****

Cooperative interactions between plant cells and tissues enable the transport of water and nutrients **(BIG IDEA 2)** via specialized structures **(BIG IDEA 4)**, providing evidence of evolutionary change in response to environmental challenges **(BIG IDEA 1)**.

Study Tip

Make a flowchart: Make a simple flowchart of the movement of water molecules from the soil through the tissues of a plant to the outside air.



Get Ready for This Chapter

In this chapter, you will learn how a plant obtains nutrients from the air and soil, and how it transports those nutrients from sites of absorption and assimilation to other parts of the plant. Before beginning this chapter, you should be able to do the following:

- Describe the basic properties of water (see Figure 3.1, Concept 3.2, and Figure 3.4).
- Describe the basic physical processes underlying membrane transport (see Figure 7.12).
- Describe the basic process of photosynthesis (see Figure 10.5).

The following questions provide a quick check of your basic knowledge in these areas.

1. The cohesiveness of water molecules is due to

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     hydrogen bonds between water molecules.
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That's correct.

Correct!If you had trouble with this question, review the following material: Hydrogen bonding is the attraction of oppositely charged ends of polar molecules, such as water, to one another.

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        their similar polarities.
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That's incorrect. Try again.

Incorrect. Review the following material: Hydrogen bonding is the attraction of oppositely charged ends of polar molecules, such as water, to one another.

pH differences on opposite ends of water molecules.

That's incorrect. Try again.

Incorrect. Review the following material: Hydrogen bonding is the attraction of oppositely charged ends of polar molecules, such as water, to one another.

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ionic bonds between water molecules.
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That's incorrect. Try again.

Incorrect. Review the following material: Hydrogen bonding is the attraction of oppositely charged ends of polar molecules, such as water, to one another.

2. The diffusion of free water across a selectively permeable membrane is called

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hydrolysis.
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That's incorrect. Try again.

Incorrect. Review the following material: Water, like all other molecules, undergoes net diffusion to an area where its free (unbound) concentration is lower. Osmosis refers to the diffusion of water across a selectively permeable membrane, resulting in the net movement of water to the side with the lower free water concentration (higher solute concentration).

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osmosis.
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That's correct.

Correct!If you had trouble with this question, review the following material: Water, like all other molecules, undergoes net diffusion to an area where its free (unbound) concentration is lower. Osmosis refers to the diffusion of water across a selectively permeable membrane, resulting in the net movement of water to the side with the lower free water concentration (higher solute concentration).

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hydration.
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That's incorrect. Try again.

Incorrect. Review the following material: Water, like all other molecules, undergoes net diffusion to an area where its free (unbound) concentration is lower. Osmosis refers to the diffusion of water across a selectively permeable membrane, resulting in the net movement of water to the side with the lower free water concentration (higher solute concentration).

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active transport.
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That's incorrect. Try again.

Incorrect. Review the following material: Water, like all other molecules, undergoes net diffusion to an area where its free (unbound) concentration is lower. Osmosis refers to the diffusion of water across a selectively permeable membrane, resulting in the net movement of water to the side with the lower free water concentration (higher solute concentration).

3. Which of the following is an end product of photosynthesis?

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carbon dioxide
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That's incorrect. Try again.

Incorrect. Review the following material: Carbon dioxide, water, and light energy are the inputs for photosynthesis. Glucose and oxygen are the end products.

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ATP
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That's incorrect. Try again.

Incorrect. Review the following material: Carbon dioxide, water, and light energy are the inputs for photosynthesis. Glucose and oxygen are the end products.

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water
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That's incorrect. Try again.

Incorrect. Review the following material: Carbon dioxide, water, and light energy are the inputs for photosynthesis. Glucose and oxygen are the end products.

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glucose
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That's correct.

Correct!If you had trouble with this question, review the following material: Carbon dioxide, water, and light energy are the inputs for photosynthesis. Glucose and oxygen are the end products.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 36.1Give examples of plant adaptations for acquiring resources.
- 36.2Distinguish between short- and long-distance transport mechanisms in a plant.
- 36.3Relate transpiration to the transport of water and minerals in a plant.
- 36.4Explain the role of stomata in regulating transpiration.
- 36.5Trace the path of sugar in a plant from source to sink.
- 36.6Describe the structure and function of the symplast.